Hence, the magnetic force at element *dl* of the wire is W_1 is

$$\vec{F} = \left(I_1 \vec{dl} \times \vec{B}_2\right) = I_1 dl \frac{\mu_\circ I_2}{2\pi r}(\hat{k} \times \hat{i})$$
$$= \frac{\mu_\circ I_1 I_2 dl}{2\pi r}\hat{j} \qquad (3.67)$$

Therefore the force on d*l* of conductor A is directed towards the conductor B. So the length d*l* is attracted towards the conductor B as shown in Figure (3.59).

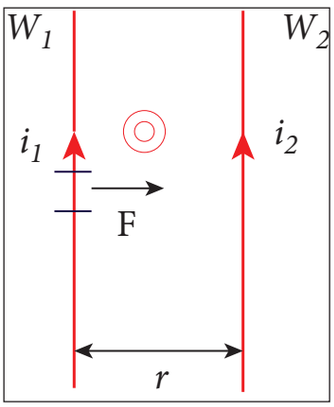


Figure 3.59 Current in both the wire are in the same direction - attracts each other

The force per unit length of the conductor A due to the conductor B is

$$\frac{\vec{F}}{l} = -\frac{\mu_\circ I_1 I_2}{2\pi r}\hat{j}$$

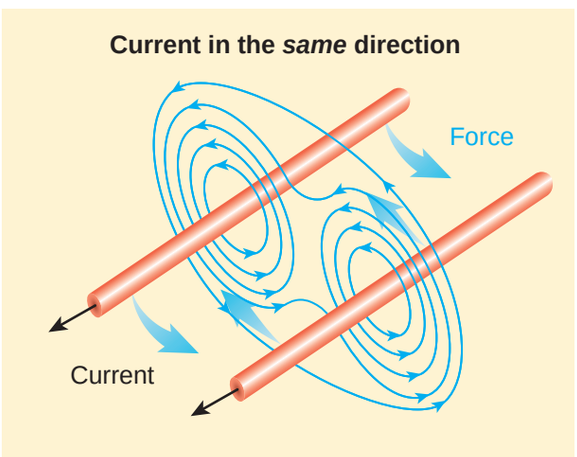


Figure 3.60 Two parallel conductors carrying current in same direction experience an attractive force

Thus the force experienced by two parallel current carrying conductors is attractive if the direction of electric current passing through them is same as shown in Figure 3.60.

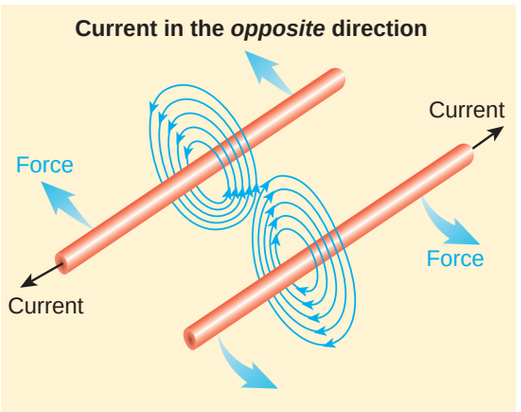


Figure 3.61 Two parallel conductors carrying current in opposite direction experience a repulsive force

Thus the force experienced by two parallel current carrying conductors is repulsive if they carry current in the opposite directions as shown in Figure 3.61.

Definition of ampère

One ampère is defined as that current when it is passed through each of the two infinitely long parallel straight conductors kept at a distance of one meter apart in vacuum causes each conductor to experience a force of 2×10^{-7} newton per meter length of conductor.

3.11 TORQUE ON A CURRENT LOOP

The force on a current carrying wire in a magnetic field is responsible for the motor operation.

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